



TATA ELECTRONICS

- **Company:** Tata Electronics
- **Type:** Semiconductor Foundry
- **Location:** Dholera, Gujarat
- **Project Cost:**
\$11 billion
(₹91,000 crore)
- **Type of Chips:** Power management IC, display drivers, microcontrollers (MCU) and high-performance computing logic
- **Industry:** Automotive, computing and data storage, wireless communication

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- **Type:** Packaging & Testing
 - **Location:** Jagiroad, Assam
 - **Project Cost:**
\$3.3 billion
(₹27,000 crore)

- **Type of Chips:** Wire Bond, Flip Chip, and a differentiated offering called Integrated Systems Packaging (ISP)
- **Industry:** Automotive (especially electric vehicles), communications, network infrastructure

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*CURRENCY CONVERSION IS APPROXIMATE

While China invested billions over two decades to build its chip manufacturing and packaging ecosystem, it missed making big investments in WFE manufacturing. Instead, it relied on WFE from the US, EU, and South Korea.

“Eight types of equipment are used for semiconductor manufacturing. Preparations have begun to make five of those eight categories. There are other categories for which India must work hard and attract more talent,” says Vaishnav.

He says a major component of lithography [the equipment used to etch electrical circuit patterns on

silicon wafers] is also being developed in India, but “the government will share details later”.

As the demand for chips is forecast to surge by 2030, semiconductor companies are building more fabs. A critical question remains: where are the workers who can operate these fabs? A joint study by the Semiconductor Industry Association of the US and Oxford Economics has projected a deficit of 67,000 technicians, computer scientists, and engineers by 2030, with a broader gap of 1.4 million such workers anticipated in the US alone. Fabs are coming up not only in the US; many countries, including India, are building their fabs.

A **S INDIA AIMS** to become a leading chip producer, the government has initiated a mission to train over 85,000 engineers. The All India Council for Technical Education (AICTE) has tailored a curriculum for a BTech in Electronics (VLSI design and technology) and a diploma in IC manufacturing.

Rajeev Chandrasekhar, Minister of State for Electronics & IT, says, “We have witnessed an impressive enrolment of almost 12,000 engineers in VLSI degree programs under the new curriculum this year and it is gratifying.” The tangible results: Micron has already recruited talent from Gujarat’s Nirma University for its ATMP unit in Sanand.

Setting up fabs is fine, but true leverage comes from creating semiconductor intellectual property (IP). An industry expert working with one of the world’s leading chip firms says, “I believe it is also a good time to take a more long-term approach in the form of strategic and advanced R&D pillars by applying the same approach that India has

adopted in manufacturing from an incentive policy and orchestration standpoint. India must also build capacity and skills and have IPs in differentiated semiconductor R&D and manufacturing areas. I can’t think of a better time to do it.”

Singapore and Taiwan are some of the countries that have adopted these templates, and India can leverage them, too. Recently, the US announced plans to invest over \$5 billion in the newly set up National Semiconductor Technology Center, a public-private consortium that will encourage semiconductor R&D. The NSTC is expected to keep the US ahead in semiconductor R&D and reduce the time and cost of commercialising new technologies, among other things.

India has already conceptualised its Bharat Semiconductor Research Centre (BSRC), which Chandrasekhar says will be a world-class partnership of academia, government, and the private sector.

Faruqui says the BSRC’s technology portfolio is even more diverse than the NSTC’s, as it will work on advanced silicon fabrication, advanced packaging R&D, compound/power semiconductor, chip design and electronic design automation.

Having lost 50 years, India is now strategically directing its efforts across all sectors of the semiconductor ecosystem. From design and skill development to ATMPs, fabs, chemicals and gases, equipment manufacturing, and R&D, India aims to secure a position among the Top 5 semiconductor players globally in the next five years.

That’s important because experts say that 30-40 years from now, nations engaged in semiconductor production will be among those calling the shots. **BT**

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